

Exploring the AKS cluster – Kubernetes side

- **metrics-server**
 - Installed as a deployment in AKS
 - Aggregator of resource usage data like CPU and memory
 - Fetches metrics from the kubelet running on nodes and exposes them in Kubernetes API server
 - Can be used by:
 - Horizontal Pod Autoscaler (HPA) – pod autoscaler based on CPU and memory
 - Vertical Pod Autoscaler (VPA) – tool that automatically adjusts or recommends limits and requests for containers in their pods
 - The metrics can be queried by cluster users via the “kubectl top” command

```
$kubectl top node
NAME                                CPU(cores)   CPU%   MEMORY(bytes)   MEMORY%
aks-nodepool1-33156667-vmss000000  82m          4%     1518Mi          33%
aks-nodepool1-33156667-vmss000001  77m          4%     1495Mi          32%
$kubectl top pod -n default
NAME                                CPU(cores)   MEMORY(bytes)
flask                               1m           43Mi
flask-app-deployment-6fbd57bfb8-km2s7 1m           43Mi
nginx                               0m           3Mi
nsenter-hzrs88                      0m           3Mi
```